MTH 202 Chapter 5: Topics In Vector Calculus

5.2 Line Integrals Worksheet

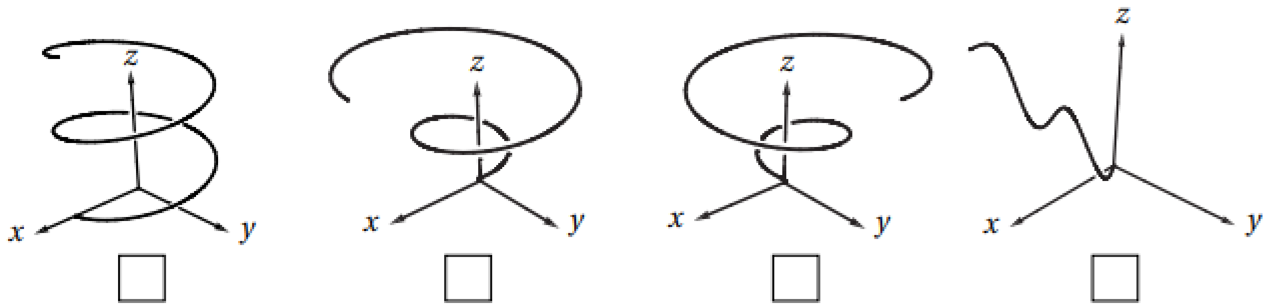
1. Let C be the curve in $\mathbb{R}^3$ parameterized by $\mathbf{r}(t) = \langle \sin(t), 2t, \cos(t) \rangle$ for $0 \le t \le \dfrac{\pi}{2}$.

a) Evaluate the integral $\int_C x^2 z \, ds$.

b) Evaluate the integral $\int_C x^2 z \, dx$.

c) Evaluate the integral $\int_C x^2 z \, dy$.

2. Mark the picture of the curve in $\mathbb{R}^3$ parameterized by $\mathbf{r}(t)=\langle t\cos(t), t\sin(t), t\rangle$ for $0 \le t \le 4\pi$.

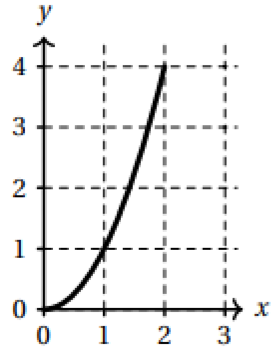


3. Let C be the curve parameterized by $\mathbf{r}(t)=\langle\cos(t), \sin(t), 2t\rangle$ for $0 \le t \le 2\pi$. Compute $\int_C z\,ds$.

4. Find the mass of a thin wire in the shape of the curve $\mathbf{r}(t)=\langle\sin(t), 2\sin(t), \sqrt{5}\cos(t)\rangle$ where $0 \le t \le \pi$, if the wire has density function $\rho(x,y,z)=x$.

5. Suppose that $\mathbf{r}:[0,2]\to\mathbb{R}^2$ is a parametric curve in the plane and that $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ have the values given in the table at left. Check the box below the picture that could be a plot of $\mathbf{r}(t)$.

t	$\mathbf{r}(t)$	$\mathbf{r}'(t)$
0	$(0,0)$	$\mathbf{i}$
1	$(1,1)$	$-\mathbf{i}$
2	$(2,4)$	$\mathbf{i}+4\mathbf{j}$



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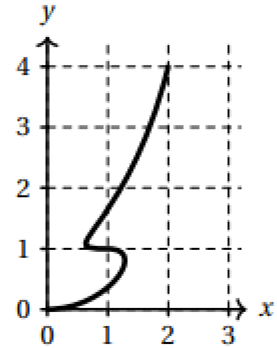

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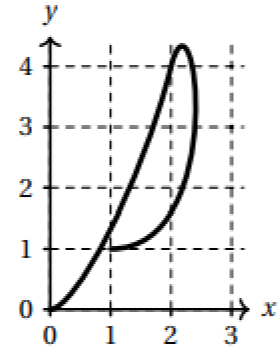

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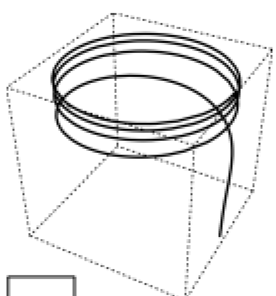
6. Let C be the oriented curve parameterized by $\mathbf{r}(t)=\langle\cos(t),\sin(t),e^t\rangle$ where $0\le t\le 8\pi$.

a) Check the box next to the picture which best matches C.



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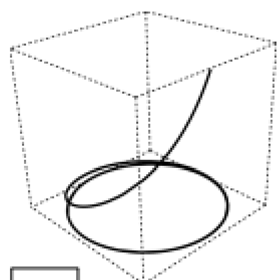

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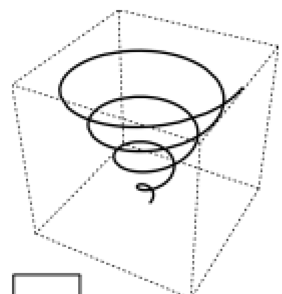

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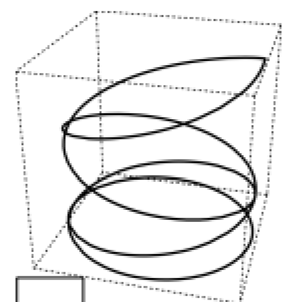

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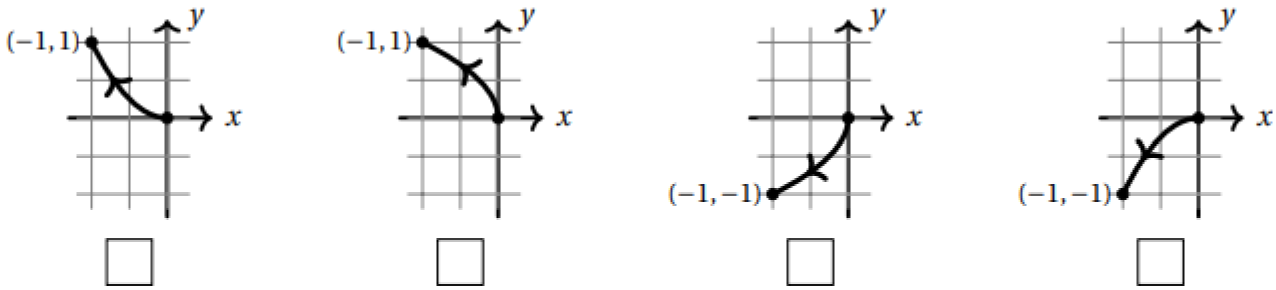
b) Calculate the line integral $\int_C z\,dz$

7. Suppose a wire can be modeled by the segment of the circle $x^2 + y^2 = 4$ from $(2,0)$ to $(0,2)$, when moving counterclockwise. If the linear density of this wire is measured by $f(x,y) = xy$ for any point on the wire, then set up an integral which represents the mass of this wire.

8. Let $f(x,y,z) = xy - xz$, and C be a curve which is the intersection of the surfaces $y = x^2 + 1$ and $z = x^2 - 1$ from $(0,1,-1)$ to $(1,2,0)$. Compute the line integral $\int_C f(x,y,z)\,ds$.

9. Let C be the curve in $\mathbb{R}^2$ parameterized by $\mathbf{r}(t) = \langle -t^2, t \rangle$ for $0 \le t \le 1$.

a) Mark the picture of C from among the choices below.



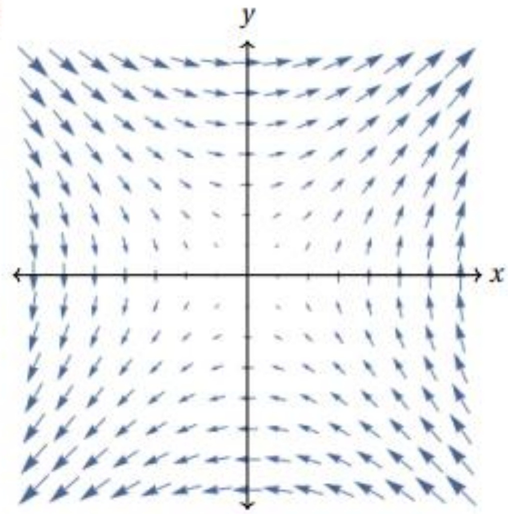
b) For the vector field $\mathbf{F} = \langle y, -x \rangle$ compute $\int_C \mathbf{F} \cdot d\mathbf{r}$.

10. Three continuous vector fields **F**,**G**, **H** on $\mathbb{R}^2$ are plotted below in the region with $-1 < x < 1$ and $-1 < y < 1$. Circle the best response for each of the following questions.

a) Which vector field represents $\vec{\nabla} f$, where $f(x,y) = xy$.

F	**G**	**H**

F

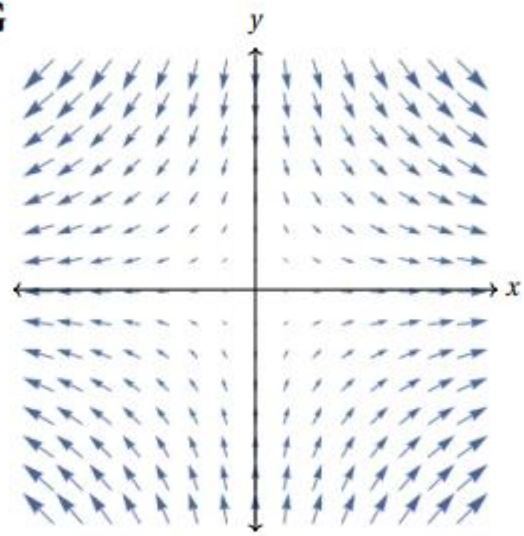


b) A particle moves along a straight line from $(-1,-1)$ to $(1,1)$. If the vector fields represent force fields, then:

The work done by **G** on the particle is…

positive	negative	zero

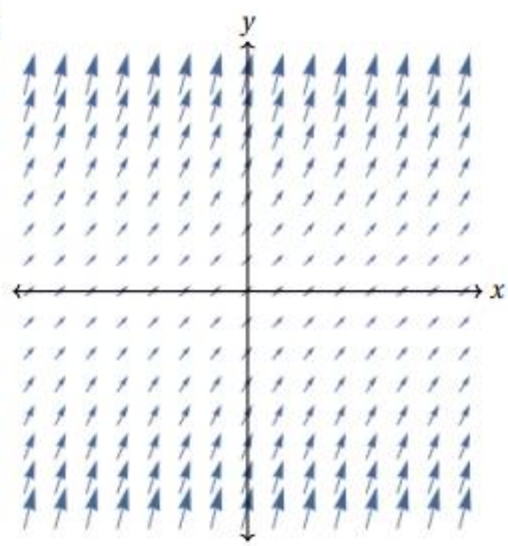
G



The work done by **H** on the particle is…

positive	negative	zero

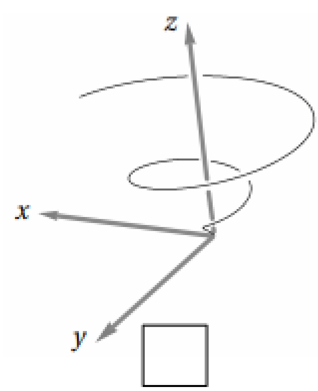
H

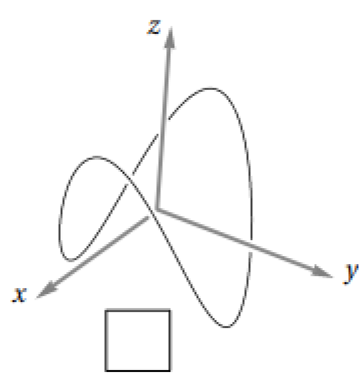


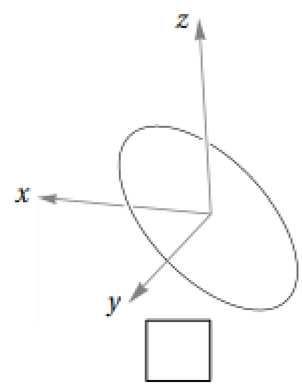
11. Find $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$ and C is the path along the circle of radius 3 starting at the point $(3,0)$ and ending at the point $(0,-3)$ while moving clockwise.

12. Consider the curve C parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), \cos(2t) \rangle$ where $0 \le t \le 2\pi$.

a) Check the box next to the correct sketch of C.







b) Find the work done if a particle travels along path $\mathbf{r}(t)$ under the force given by the vector field $\mathbf{F}(x,y,z) = \langle -2y, 2x, 0 \rangle$.

13. Let C be the helix parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ where $0 \le t \le 2\pi$.

a) Let $\mathbf{F}(x, y, z)$ be the vector field defined by $\mathbf{F}(x, y, z) = \langle x, y, 0 \rangle$. Determine $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is oriented in the direction from $(1,0,0)$ to $(1,0,2\pi)$.

b) Now let $\mathbf{F}(x, y, z) = \left\langle ye^{x^2+y^2}, \ln\left(x^2 + y^2\right) - x, 1 \right\rangle$. Determine $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is given the opposite orientation.